

GRADE 6 • LIFE SCIENCE

Vertebrates and Invertebrates

Discover how animals are classified



What Are Vertebrates and Invertebrates?



VERTEBRATE

An animal with a backbone
or vertebral column



INVERTEBRATE

An animal without
a backbone

Look around you — which animals might have backbones?

Meet the Five Vertebrate Groups

Their backbones support the body and protect the spinal cord.



FISH

AMPHIBIANS

REPTILES

BIRDS

MAMMALS

One backbone family • many different body plans

WATER EXPLORERS

Fish

- Live mostly in water
- Breathe using gills
- Most have fins and scales

Examples • Tilapia • Tuna



TWO-WORLD LIVES

Amphibians

- Begin life mostly in water
- Adults may live on land
- Moist, smooth skin

Examples • Frog • Salamander



SCALED SURVIVORS

Reptiles

- Dry skin covered in scales
- Cold-blooded animals
- Most lay eggs on land

Examples • Snake • Turtle • Lizard



FEATHERED VERTEBRATES

Birds

- Feathers cover the body
- Wings and a beak
- Lay hard-shelled eggs

Examples • Eagle • Chicken



WARM & CARING

Mammals

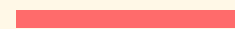
- Hair or fur
- Warm-blooded animals
- Mothers produce milk

Examples • Dog • Cat • Human



What Are Invertebrates?

Animals without a backbone.
They make up most animal species on Earth!



NO BACKBONE • AMAZING
VARIETY



Six Amazing Invertebrates

BUTTERFLY

Insect • six legs and wings

SPIDER

Arachnid • eight legs

EARTHWORM

Worm • soft, long body

SNAIL

Mollusk • soft body and shell

JELLYFISH

Cnidarian • jelly-like body

OCTOPUS

Mollusk • eight arms

Vertebrates vs. Invertebrates

VERTEBRATES

- ✓ Have a backbone
- ✓ Internal skeleton
- ✓ Fish • amphibians • reptiles
birds • mammals

INVERTEBRATES

- ✗ No backbone
- ✓ May have shells or exoskeletons
- ✓ Insects • worms • mollusks
spiders • many more

BACKBONE = VERTEBRATE • NO BACKBONE = INVERTEBRATE

Activity 1 • Vertebrate or Invertebrate?

Say your answer first — then check the reveal!

1

DOG

Vertebrate

2

BUTTERFLY

Invertebrate

3

FISH

Vertebrate

4

SPIDER

Invertebrate

5

CHICKEN

Vertebrate

Activity 2 • Which Vertebrate Group?

Match each animal to its group.

?

Tilapia

FISH

?

Frog

AMPHIBIAN

?

Snake

REPTILE

?

Eagle

BIRD

?

Cat

MAMMAL

Activity 3 • Sort the Animals

Point, call, or drag each card to the correct side!

VERTEBRATES

Dog

Fish

Frog

Butterfly

Spider

Earthworm

INVERTEBRATES

Chicken

Turtle

Snail

Octopus

Assessment Quiz

Choose A, B, C, or D.

1 What is a vertebrate?

A No legs B Has a backbone C Has wings D Lives in water

2 Which animal is a vertebrate?

A Spider B Butterfly C Fish D Snail

3 Which group does a frog belong to?

A Fish B Reptile C Amphibian D Mammal

4 Which animal is an invertebrate?

A Dog B Eagle C Turtle D Butterfly

5 What is the main difference?

A Size B Backbone C Color D Habitat

Answer Key

Q1

B — An animal with a backbone

Q2

C — Fish

Q3

C — Amphibian

Q4

D — Butterfly

Q5

B — Vertebrates have a backbone; invertebrates do not

Key Takeaways

01

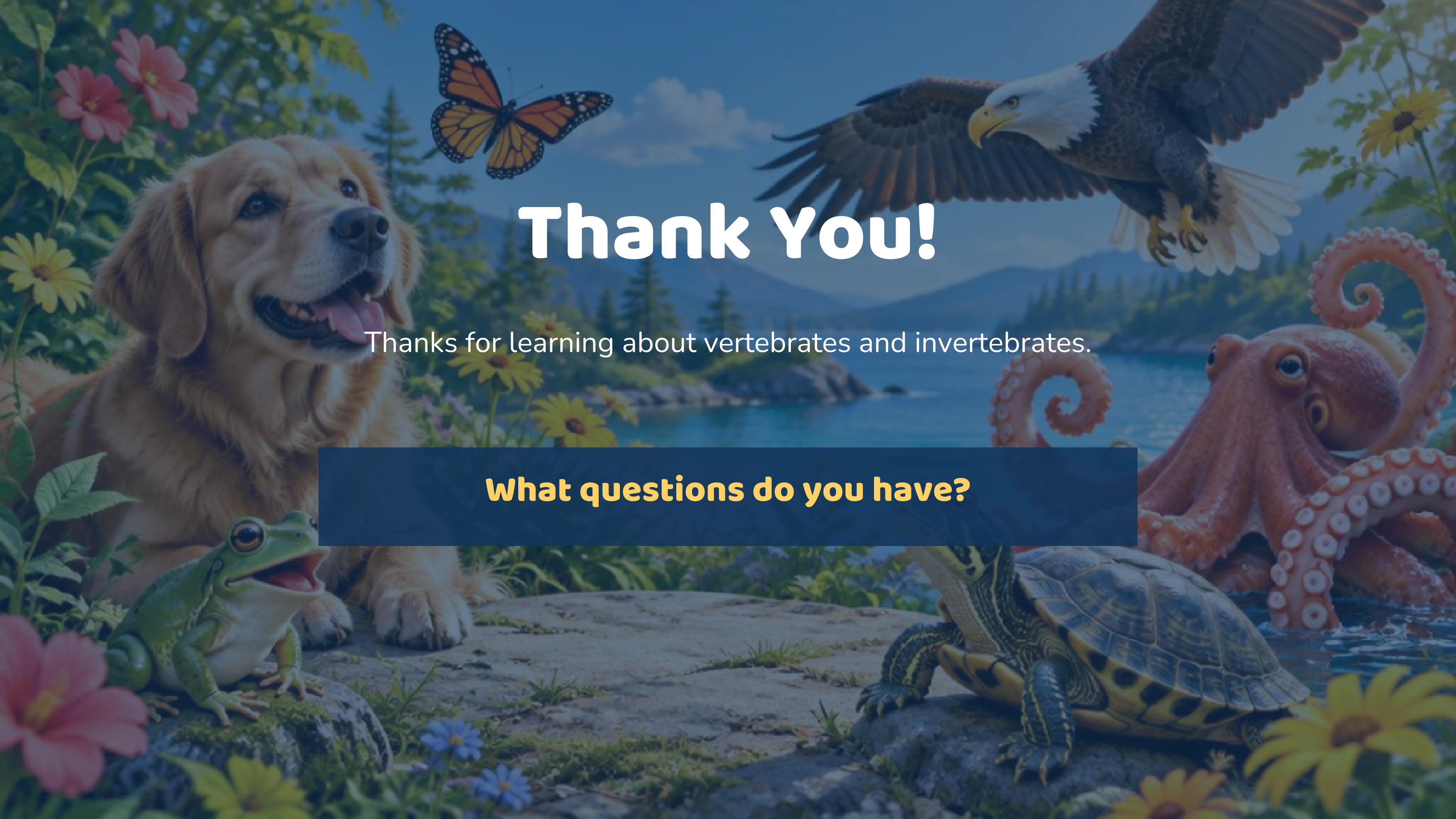
Vertebrates have backbones: fish, amphibians, reptiles, birds, and mammals.

02

Invertebrates lack backbones and include insects, worms, mollusks, spiders, and more.

03

Observable body features help us classify animals — stay curious about life around you!



Thank You!

Thanks for learning about vertebrates and invertebrates.

What questions do you have?